

Lab Sheet-7

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```
import os

import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.preprocessing import StandardScaler

from sklearn.cluster import KMeans

from sklearn.metrics import silhouette_score

# 0. Setup Output Directory

output_dir = "output_images"

os.makedirs(output_dir, exist_ok=True)

# 1. Load Dataset

DATA_URL = "https://raw.githubusercontent.com/sharmaroshan/Clustering-of-Mall-Customers/master/Mall_Customers.csv"

df = pd.read_csv(DATA_URL)

# Clean column headers

df.columns = df.columns.str.strip().str.replace(' ', '_').str.replace('(', '').str.replace(')', '')

# Select features & scale

feature_cols = ['Annual_Income_k$', 'Spending_Score_1-100']

X = df[feature_cols].copy()

scaler = StandardScaler()

X_scaled = scaler.fit_transform(X)

# 2. Elbow & Silhouette Curves

wcss = []

k_values = range(2, 11)

silhouette_scores = []

for k in k_values:
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km = KMeans(n_clusters=k, init='k-means++', n_init=10, random_state=42)

labels = km.fit_predict(X_scaled)

wcss.append(km.inertia_)

silhouette_scores.append(silhouette_score(X_scaled, labels))

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Elbow Curve

axes[0].plot(list(k_values), wcss, marker='o', color='#1f77b4', linewidth=2)

axes[0].set_title('Elbow Method (WCSS vs. K)')

axes[0].set_xlabel('Number of Clusters (K)')

axes[0].set_ylabel('Inertia (WCSS)')

axes[0].grid(True, linestyle='--', alpha=0.6)

# Silhouette Curve

axes[1].plot(list(k_values), silhouette_scores, marker='s', color='#2ca02c', linewidth=2)

axes[1].set_title('Silhouette Score vs. K')

axes[1].set_xlabel('Number of Clusters (K)')

axes[1].set_ylabel('Silhouette Score')

axes[1].grid(True, linestyle='--', alpha=0.6)

plt.tight_layout()

# Save Image 1

fig.savefig(os.path.join(output_dir, "elbow_silhouette_analysis.png"), dpi=300,
bbox_inches='tight')

print(f"Saved: {os.path.join(output_dir, 'elbow_silhouette_analysis.png')}")

plt.show()

# 3. Fit Model & Plot Clusters

optimal_k = 5

kmeans = KMeans(n_clusters=optimal_k, init='k-means++', n_init=10, random_state=42)

df['Cluster'] = kmeans.fit_predict(X_scaled)

centroids_original = scaler.inverse_transform(kmeans.cluster_centers_)

```

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plt.figure(figsize=(9, 6))

palette = ['#e41a1c', '#377eb8', '#4daf4a', '#984ea3', '#ff7f00']

sns.scatterplot(
    data=df,
    x='Annual_Income_k$',
    y='Spending_Score_1-100',
    hue='Cluster',
    palette=palette,
    s=70,
    alpha=0.9
)

# Centroids
plt.scatter(
    centroids_original[:, 0],
    centroids_original[:, 1],
    s=250,
    c='black',
    marker='X',
    edgecolor='white',
    linewidth=1.5,
    label='Centroids'
)

plt.title('Customer Segments (K-Means Clustering, K=5)', fontsize=13)
plt.xlabel('Annual Income (k$)')
plt.ylabel('Spending Score (1-100)')
plt.legend(bbox_to_anchor=(1.02, 1), loc='upper left')
plt.grid(True, linestyle='--', alpha=0.5)
plt.tight_layout()

```

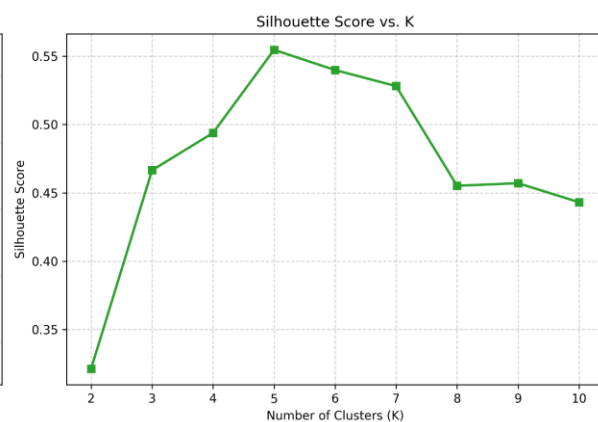
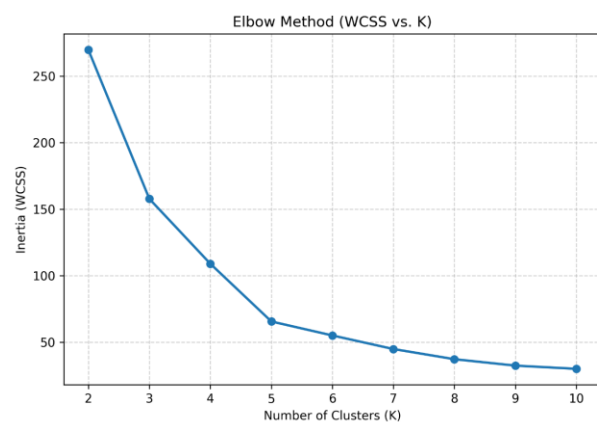
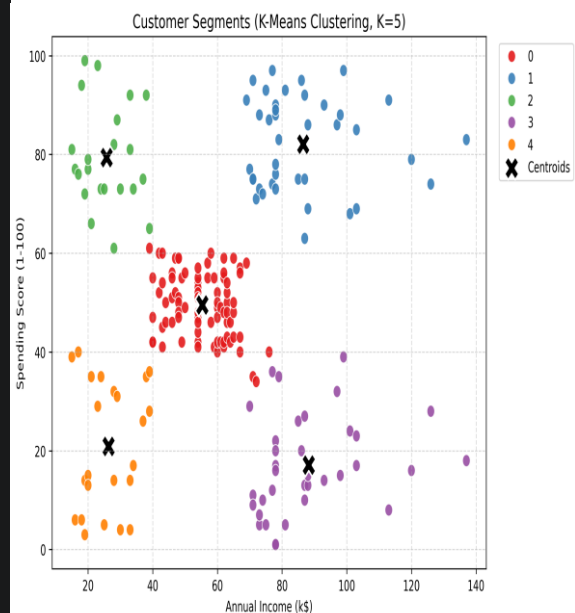
```
# Save Image 2
```

```
plt.savefig(os.path.join(output_dir, "customer_segments_clusters.png"), dpi=300,  
bbox_inches='tight')
```

```
print(f"Saved: {os.path.join(output_dir, 'customer_segments_clusters.png')}")
```

```
plt.show()
```

```
Dataset loaded successfully.  
Dataset Shape: (200, 5)  
|  
--- First 5 Rows ---  
| CustomerID  Genre  Age  Annual Income (k$)  Spending Score (1-100)  
0           1   Male   19              15              39  
1           2   Male   21              15              81  
2           3  Female  20              16               6  
3           4  Female  23              16              77  
4           5  Female  31              17              40  
  
--- Missing Value Check ---  
CustomerID      0  
Genre            0  
Age              0  
Annual_Income_k$ 0  
Spending_Score_1-100 0  
dtype: int64  
  
--- Final Cluster Centroids & Profiles ---  
Cluster  Annual_Income_k$  Spending_Score_1-100  Customer_Count  
0         55.296296         49.518519             81  
1         86.538462         82.128205             39  
2         25.727273         79.363636             22  
3         88.200000         17.114286             35  
4         26.304348         20.913043             23
```



Labs Sheet Questions

1. What is clustering? How does it differ from classification?

- **Clustering:** An unsupervised learning technique used to partition unlabeled data into natural groupings based on structural or geometric similarities.
- **Difference from Classification:**
 - *Clustering (Unsupervised):* Works without predefined target classes or ground-truth labels. The model discovers patterns purely from feature similarities.
 - *Classification (Supervised):* Trains on labeled input-output pairs to map features to predefined target categories.

2. Explain the working principle of the K-Means Clustering algorithm.

K-Means partitions data into K distinct clusters by minimizing the Within-Cluster Sum of Squares (Inertia):

1. **Initialization:** Select K initial points as cluster centroids (often using K -means++ to space them apart).
2. **Assignment:** Assign each sample to its nearest centroid based on Euclidean distance.
3. **Update:** Recalculate each centroid as the arithmetic mean of all data points assigned to that cluster.
4. **Convergence:** Repeat the assignment and update steps iteratively until centroids remain stationary or a maximum iteration limit is reached.

3. Why is feature scaling important before applying K-Means clustering?

K-Means computes pairwise distances (Euclidean distance) across feature dimensions. If features have different scales (e.g., Annual Income in thousands versus Spending Score scaled from 1 to 100), the feature with larger numerical values will dominate the distance metric. Standardization ensures all features contribute equally to the distance calculation.

4. What is the Elbow Method, and how does it help determine the optimal number of clusters?

The Elbow Method plots the Within-Cluster Sum of Squares (WCSS / Inertia) against a range of cluster counts (K). As K increases, WCSS decreases monotonically. The "elbow" point represents the inflection where the rate of decrease drops sharply, indicating the optimal balance between cluster compactness and model complexity.

5. What is the Silhouette Score? How is it used to evaluate clustering performance?

The Silhouette Score measures how well-separated and cohesive clusters are. For an individual sample i :

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

- $a(i)$: Mean intra-cluster distance to all other points in the same cluster.
- $b(i)$: Mean nearest-cluster distance to points in the closest neighboring cluster.

Values range from -1 to $+1$:

- **Close to $+1$** : Well-separated, dense clusters.
- **Around 0** : Overlapping cluster boundaries.
- **Negative**: Samples may be assigned to the wrong cluster.

6. Why is K-Means considered an unsupervised machine learning algorithm?

It operates strictly on unlabeled input features without target variables or ground-truth class assignments. The algorithm derives groupings autonomously based on feature geometry and proximity rather than learning from labeled examples.

7. Mention any four real-world applications of customer segmentation.

- **Targeted Marketing Campaigns**: Delivering custom promotions tailored to specific spending tiers.
- **Customer Churn Prevention**: Identifying declining activity patterns in at-risk segments to deploy retention offers.
- **Personalized Product Recommendations**: Tailoring catalog feeds and bundle offerings based on cluster purchasing habits.
- **Dynamic Pricing & Credit Limits**: Assigning credit card limits, financing options, or promotional pricing based on spending behavior and income levels.

8. What are the limitations of the K-Means algorithm?

- **Fixed K Requirement**: Requires defining the number of clusters beforehand.
- **Spherical Cluster Bias**: Assumes clusters are spherical and isotropic; fails to detect complex, non-linear, or elongated cluster shapes.
- **Sensitivity to Outliers**: Because cluster centers are calculated as means, extreme values distort centroid coordinates.
- **Sensitivity to Initialization**: Random centroid placement can lead to convergence at suboptimal local minima.

9. How can businesses use customer segmentation to improve marketing and customer retention?

- **Targeted Campaigns:** By identifying distinct groups (e.g., high-income/high-spending vs. budget spenders), companies avoid generic ads and run high-conversion campaigns tailored to each group's budget.
- **VIP Retention:** Dedicated loyalty perks, dedicated managers, and early access are allocated to top-tier spenders to maximize lifetime value.
- **Re-engagement:** Price-sensitive or low-engagement customers receive discount codes or flexible payment models (e.g., BNPL) before churning.

10. Compare K-Means Clustering with Hierarchical Clustering based on their working principles and applications.

Aspect	K-Means Clustering PDF	Hierarchical Clustering (Agglomerative) PDF
Working Principle	Partitions data by iteratively updating K centroids using Euclidean distance.	Builds a bottom-up tree by sequentially merging the closest pairs of clusters.
Cluster Count (K)	Must be chosen prior to running the model.	Does not require an initial K ; the tree can be sliced at any height.
Scalability	$\mathcal{O}(n \cdot K \cdot I \cdot d)$; scalable to large datasets.	$\mathcal{O}(n^2 \log n)$ to $\mathcal{O}(n^3)$; computationally prohibitive for large datasets.
Output Representation	Discrete cluster assignments with centroid coordinates.	A dendrogram showing nested sub-clusters and hierarchical structure.
Common Uses	Customer segmentation, document clustering, and large-scale tabular analysis.	Gene expression analysis, taxonomic classifications, and exploratory studies on small datasets.